

Business Problem Statement

A leading retail company seeks to gain deeper insights into customer shopping behavior to enhance sales performance, improve customer satisfaction, and strengthen long-term customer loyalty. Recent shifts in purchasing patterns across customer demographics, product categories, and sales channels (online and offline) have created a need for data-driven decision-making.

The management team aims to identify the key factors influencing consumer purchasing decisions and repeat buying behavior. Specifically, they want to understand how variables such as discounts, customer reviews, seasonal trends, payment methods, and shopping channels impact customer engagement and sales outcomes.

Business Question

How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

Project Objectives

Identify customer purchasing trends and behavioral patterns
Understand factors driving customer satisfaction and loyalty
Segment customers based on demographics and purchasing behavior
Evaluate the impact of discounts, reviews, and seasonal trends on sales
Compare online and offline shopping behaviors
Improve marketing effectiveness and product strategy decisions

Deliverables

1. Data Preparation & Modeling (Python)

Clean, preprocess, and transform the raw dataset for analysis
Handle missing values, duplicates, and inconsistent data formats
Perform feature engineering where necessary
Prepare datasets suitable for analytical and visualization tasks

2. Data Analysis (SQL)

Design and organize the dataset into a structured relational format
Simulate business transactions and customer interactions
Extract insights related to customer segmentation, sales performance, loyalty, and purchase drivers

3. Visualization & Insights (Power BI)

Develop interactive dashboards for customer behavior and sales analysis
Highlight trends, product performance, and marketing effectiveness
Enable data-driven decision-making through visual storytelling

4. Report & Presentation

Summarize methodology, findings, and recommendationsPrepare a stakeholder-friendly presentation with visual insights

5. GitHub Repository

Include Python scripts, SQL queries, dashboard files, and documentationMaintain a clean and well-structured repository for collaboration